Unbalanced observers and the LCA restriction

What we assume and what we could assume instead

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The problem: unbalanced observers have no trajectory label

A trajectory $\underline{d}_i$ is a **full-wave** sequence of urban/rural status

- balanced observers in IDN: observed in all 5 waves $\Rightarrow$ full trajectory
- unbalanced observers: missing at least one wave $\Rightarrow$ no well-defined $\underline{d}_i$

Yet unbalanced observers are 88–96% of the sample

Dropping them means dropping most of the data

What we currently do: pool them into one extra cell

Think of it as a new cell alongside the balanced trajectory cells:

	Rural consumption mean	Average urban return
Balanced cell $\underline{d} \in \mathcal{D}$	$\mu_{\underline{d}}$	$\Delta_{\underline{d}}$
Pooled unbalanced cell	μ_{unb}	Δ_{unb}

In the regression, this is implemented by adding an unbalanced dummy U_i and its interaction with the urban indicator D_{it}

Their coefficients *are* μ_{unb} and $\Delta_{\mathrm{unb}} - \Delta_{\underline{d}_0}$ — just relabeled

What LCA says about balanced cells

The LCA restriction (A5) at the individual level: $\Delta_i = \beta + \phi\,\theta_i$

Averaged within a trajectory cell, this gives the **cell-level LCA**:

$$\Delta_{\underline{d}} = \Delta_{\underline{d}_0} + \phi\,(\mu_{\underline{d}} - \mu_{\underline{d}_0}) \qquad \text{for every } \underline{d} \in \mathcal{D}_S$$

Every balanced switcher cell has one free parameter ($\mu_{\underline{d}}$)
The return $\Delta_{\underline{d}}$ is determined by $\mu_{\underline{d}}$ through LCA

This is what lets us extrapolate returns to non-migrants (d_N, d_T) along the LCA line

What we're *not* imposing on the unbalanced cell

The pooled unbalanced cell, in our current spec, has two free parameters:

$$\mu_{\mathrm{unb}} \quad \text{and} \quad \Delta_{\mathrm{unb}}$$

They are estimated independently. Nothing in the current spec ties them together.

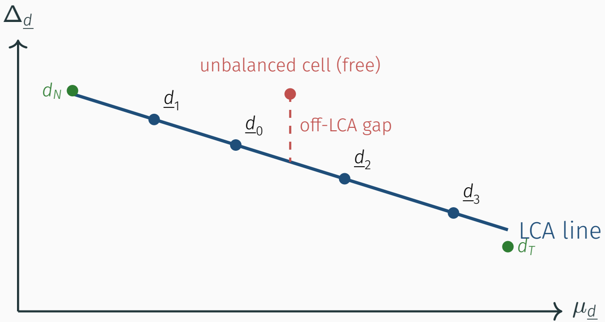
If LCA held on the unbalanced cell too, they would satisfy

$$\Delta_{\mathrm{unb}} = \Delta_{\underline{d}_0} + \phi\left(\mu_{\mathrm{unb}} - \mu_{\underline{d}_0}\right)$$

the same line that every switcher cell lies on

We are not imposing this. We let the data fit Δ_{unb} freely.

The LCA line, in a picture



Balanced switchers $\underline{d}_0, \underline{d}_1, \ldots$ lie on the LCA line by construction. Non-migrants d_N, d_T are extrapolated to the line. The unbalanced pooled cell sits wherever the data puts it — potentially off the line.

Option 1 (current): leave Δ_{unb} unconstrained

Parameters for the unbalanced cell: μ_{unb} and Δ_{unb} (two free)

Moment conditions: separate moments pin each one down

Interpretation: we let the data fit the unbalanced-cell return without assuming LCA applies there

Pro. Robust if LCA fails on the unbalanced cell (e.g., if attrition selects on θ_i in a way that breaks the LCA line)

Con. Leaves information on the table. Unbalanced observers don't help identify ϕ.

This is what's in the paper now

Option 2 (cheap swap): impose LCA on the unbalanced cell

Replace the free return with the LCA-implied value:

$$\Delta_{\mathrm{unb}} \equiv \Delta_{\underline{d}_0} + \phi\,(\mu_{\mathrm{unb}} - \mu_{\underline{d}_0})$$

Now only μ_{unb} is a free parameter for the unbalanced cell

The return Δ_{unb} is determined by μ_{unb} and ϕ, exactly as for every balanced switcher cell

What changed. One fewer free parameter. The unbalanced cell becomes just another point on the LCA line, with mean μ_{unb}.

What's new. Unbalanced observers now contribute to identifying ϕ through the same LCA restriction as every switcher cell.

Why this is cheap and principled

Cheap

- one constraint added to the GMM spec: $\Delta_{\text{unb}} = \Delta_{\underline{d}_0} + \phi(\mu_{\text{unb}} - \mu_{\underline{d}_0})$
- no new dummies, no rewriting `handle_trajectory_groups`
- re-estimation on three countries, one afternoon

Principled

- LCA (A5) is assumed to hold at the individual level for *everyone*
- if it holds for balanced observers, it should hold for unbalanced observers too — why would a missing wave break a structural restriction?
- imposing LCA on the unbalanced cell is just being consistent with what we already claim

The restriction is testable

Under the current (free-Δ_{unb}) specification, compute both sides:

- **Left side:** $\hat{\Delta_{\mathrm{unb}}}$ from the GMM output
- **Right side:** $\hat{\Delta_{\underline{d}_0}} + \hat{\phi}(\hat{\mu_{\mathrm{unb}}} - \hat{\mu_{\underline{d}_0}})$ from the other GMM estimates

Three possible outcomes:

- they agree $\Rightarrow$ LCA holds for unbalanced observers; imposing it costs nothing
- they disagree slightly $\Rightarrow$ LCA approximately holds; imposing it trades a bit of bias for precision
- they disagree a lot $\Rightarrow$ interesting finding — LCA fails on the unbalanced cell, suggesting attrition selects on comparative advantage

In all three cases, the check gives us information we don't currently have

Why we didn't do this initially (speculation)

The current specification came from treating unbalanced observers as a nuisance rather than as a cell:

- the goal was to keep them in the regression so that covariate coefficients γ are estimated from the full sample
- their urban/rural differential was not the object of interest
- the unbalanced dummy and its urban interaction got added as "controls," not as structural parameters

But treating the unbalanced group as a trajectory cell reveals:

- there is a well-defined μ_{unb}: the rural consumption mean in the unbalanced cell
- there is a well-defined Δ_{unb}: its average urban return
- and the model already says these should be LCA-linked

Three options summarized

	Option 1 (now)	Option 2	Option 3
Unbalanced cell	1 pooled cell	1 pooled cell	several cells by observed type
Parameters	$\mu_{\mathrm{unb}}, \Delta_{\mathrm{unb}}$ free	only μ_{unb} free	$\mu^{(k)}_{\mathrm{unb}}$, each LCA-linked
LCA imposed?	no	yes	yes
Helps identify ϕ?	no	yes, one cell	yes, several cells
Effort	zero	half a day	one week

Recommendation: Option 2 is the right next step. Cheap, structurally consistent with A5, and the testable-restriction check is informative either way. Option 3 only if Option 2 reveals something interesting.

Takeaway

The current approach treats the unbalanced pooled cell as exempt from LCA.

Its rural consumption mean μ_{unb} and its average urban return Δ_{unb} are estimated as two independent free parameters.

Imposing LCA on the unbalanced cell means: replacing the free return with the LCA-implied return $\Delta_{\mathrm{unb}} = \Delta_{\underline{d}_0} + \phi(\mu_{\mathrm{unb}} - \mu_{\underline{d}_0})$.

Cost: one constraint in the GMM spec.
Benefit: one more cell contributing to $\hat{\phi}$, one less free parameter, internal consistency with A5, and a new testable restriction as diagnostic.